

Ruben Valenzuela

Professor Viswanatha Rao

Deep learning AI

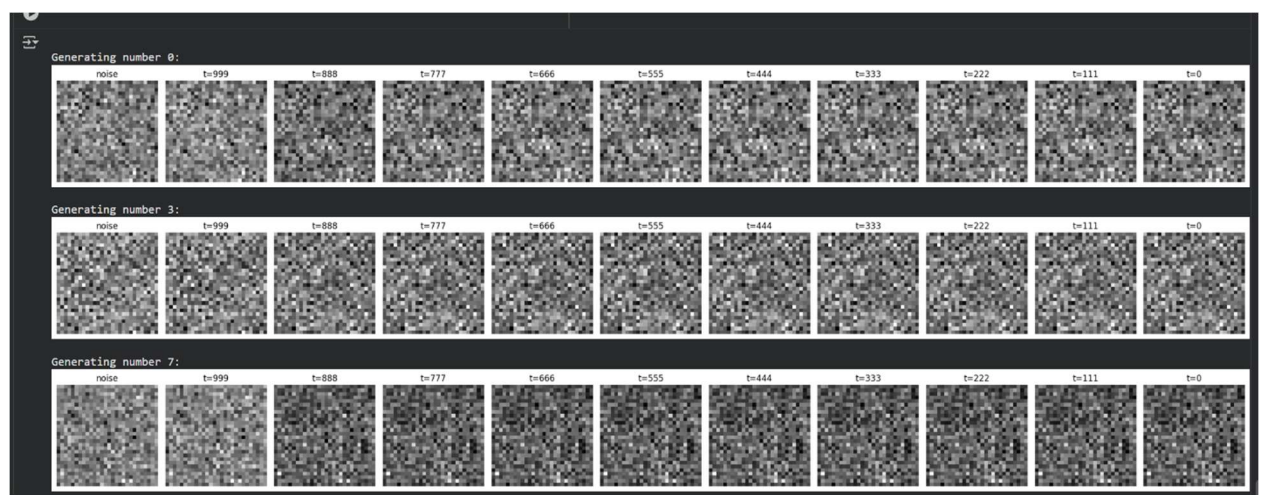
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Midterm

1. Introduction

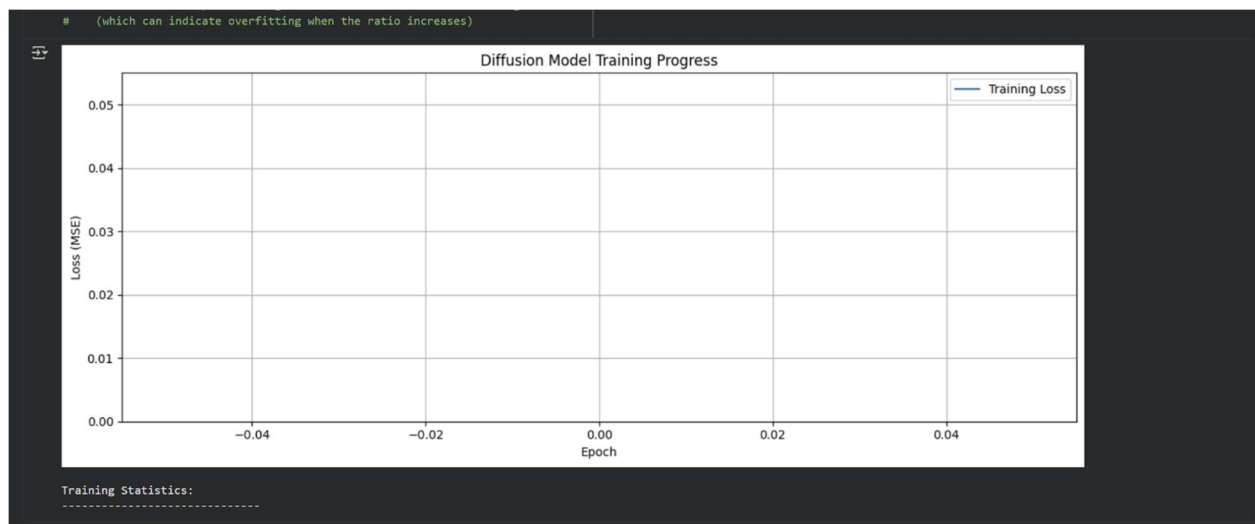
This report presents the analysis and experimental results obtained during the diffusion model lab assignment. The goal was to train and evaluate a conditional diffusion model using the MNIST dataset to generate handwritten digits. The process included forward and reverse diffusion visualization, U-Net implementation, loss monitoring, and evaluation of generated samples using CLIP for semantic quality assessment. The objective was not only to replicate results but to understand how diffusion models learn to transform noise into meaningful structure.

2. Understanding Diffusion



During the forward diffusion process, Gaussian noise is added progressively to the original MNIST digits. This gradual corruption transforms a clear digit into complete noise over many timesteps. By adding noise slowly, the reverse process becomes learnable: instead of reconstructing an image from pure noise in one step, the model learns to remove small amounts of noise incrementally. This improves training stability and convergence.

In the visualization, the digit becomes recognizable again around **40–50%** into the denoising process. Simpler digits such as **0** or **1** appear earlier, while complex digits like **8** or **9** emerge later. This demonstrates that diffusion models build structure gradually from coarse to fine details.



3. Model Architecture

The **U-Net architecture** was selected for its symmetry and multi-scale feature extraction capabilities. It consists of an encoder that compresses spatial information and a decoder that reconstructs it. **Skip connections** between encoder and decoder layers preserve high-frequency details lost during downsampling, resulting in sharper and more accurate outputs.

In this project, the model was conditioned on **class labels** using one-hot encoded vectors. These embeddings were added to the model's intermediate layers, guiding the network to generate a specific digit. A binary **mask** was also used to enable or disable class conditioning when needed.



4. Training Analysis

The **loss function** used was Mean Squared Error (MSE), which measures how accurately the model predicts the added noise at each timestep. A lower loss corresponds to better denoising capability. During early epochs, generated samples were unrecognizable blobs; as training progressed, digits became structured and visually distinct.

Time embeddings were crucial in helping the network understand how much noise remained at each step. Without them, the model would treat all steps as equal and fail to adapt its denoising behavior.



5. CLIP Evaluation

The **CLIP model** evaluated the generated digits by comparing them against text prompts such as “a handwritten number 3” or “a blurry number.” Higher similarity scores with the correct prompts indicated higher visual quality and legibility.

Digits like **0**, **1**, and **7** achieved the **highest CLIP scores** due to their simple geometric structure. Conversely, digits **5**, **8**, and **9** showed lower scores as they involve more complex curves and are prone to ambiguity.

To improve generation quality, CLIP can be used for **guided sampling or post-selection**. For example, generating multiple candidates and selecting the top samples based on CLIP scores ensures better overall clarity.



6. Practical Applications

Diffusion models like this one have broad real-world applications, such as data augmentation for OCR systems, restoration of corrupted images, and educational visualization of generative processes.

However, this specific implementation is limited by **low resolution**, absence of attention mechanisms, and a small model capacity.

Future improvements could include:

1. Adding **self-attention layers** for better spatial understanding.
2. Implementing **Exponential Moving Average (EMA)** weights for stability.
3. Experimenting with **classifier-free guidance** for more controlled and realistic outputs.

7. Reflection

Through this experiment, I developed a deeper understanding of how diffusion models synthesize structure from noise. While initial outputs were almost pure static, it was fascinating to observe order emerging gradually. I also recognized the power of CLIP as a cross-modal

evaluator of image quality. Despite training challenges, this exercise demonstrated the robustness and interpretability of diffusion-based generative models.